

FACULTY OF ENGINEERING AND TECHNOLOGY

GROUP 11 github link:

<https://github.com/emmanuelkiwanuka053-design/Group11.git>

GROUP ELEVEN		
NAME	REG_NO	Github links
AUJO ESTHER NORAH	BU/UG/2025/4440	https://github.com/Norah17-lab/MATLAB
AMONGIN CYNTHIA PATIENCE	BU/UP/2025/2105	https://github.com/Amongin-Cynthia/Amongin-Cynthia-Patience
OKOTH JOWEL	BU/UP/2025/3274	https://github.com/OkothJowel/Okoth-Jowel.git
LUWAGA EMMANUEL KIWANUKA	BU/UP/2025/2125	https://github.com/luwagaemmanuel991/Luwaga-Emmanuel-kiwanuka.git
NAGALIRA PAUL	BU/UP/2025/4627	https://github.com/mayekujoshua/NAGALIRA-PAUL

NO.1A

%Students in my class

Name=["AKITENG DEBORAH";"AKWIR MERCY LEAN";"AUMA PRETTY";"ELEPOT ERASMUS WANDERA";"OPUS JONAH";"KINOBE SOLOMON JAMES";"BWIRE ELVIS";"ACHENG GIFT OLIVER";"OKOU LAWYERS";"OLINGA JONATHAN";"AINOMUGISHA PROMISE";"NAMUGAYA EDITH";"EMORUT PIUS OJILONG";"WANGO LAWRENCE";"MUGISA JOHN";"AHURRA ARTHUR ELLY";"AMONGIN CYNTHIA PATIENCE";"MUWAYIRE GASTER";"AWIDI VANNESSA";"KABAGYENYI GIFT JOSEPHINE";"LAPOBO DORCUS";"AUJO ESTHER NORAH";"ANYANZO GERALD";"APIO REBECCA";"OGWANG SAMSON";"OLILE RICKY IGNATIUS";"DUKU FRANCIS";"KAMULI DAVID LAWRENCE";"OLWENY EMMANUEL";"NAMUBIRU HADIJAH";"MASETTE TIMOTHY";"OLEKE SOLOMON";"TURINAWA LUCKY";"ZAZIO NESSER";"MUSIIMENTA DANIEL";"ACULERA GRACE";"AKAMPA ANTHONY";"MORUNYANG RICHARD";"NABUKALU ANGEL";"OLIPOTO MISACH";"OGWEL BRIAN";"KUJO WINNIE";"MUSASIZI DAN";"KABUYE RAYHAAN";"LUMALA MUBARACKAH MULUMBA";"OBONYO JOEL";"NABUKENYA LATIFAH MATOVU";"ACHAM PRISCA SOPHIE";"OKOTH JOWEL";"RUTAYO HERBERT KARTUSI";"OKELLO ISAAC DANIEL";"NAKINTU SOPHIA";"OLUPOT RAPHEAL";"NAMUTAMBA JONITAH OBIRA";"NAGALIRA PAUL";"ATUKWATSE BRUCE";"ACHOLA ELIZABETH";"LAKER MARY ANGEL";"AJALO TRACY PATRICIA";"OMOOKA ARNOLD PETER";"AINEMBABAZI MERCY";"ODONG FREDRICK ARNOLDS";"OGOORO PATRICIA";"KYESIMIRA THEODORE";"NAKATO KEVIN";"ADONGO ESTHER";"KALISA RONALD";"OLARA EMMANUEL GEORGE";"SANDE CLINTON";"MUKAMA SAMUEL";"ORIOKOT ABRAHAM";"MPUUGA MARK";"AKAREUT FLORENCE";"LUWAGA EMMANUEL KIWANUKA";"ADONG VIVAN";"ANDIRU EMILLY";"KALIBBALA MARIAM";"NAMUKUYE JANE SHERINA";"ROMOI ABEL KARENGET";"ASIO PRISCILLA JOAN";"BIRIBAWA JULIET";"ALUM PROSSY FRANCIS";"ONAPA BRIAN";"MBYEMEIRE FRED";"OKELLO INNOCENT";"CHEMONGES RONNEY";"NAMBAYO PATRICIAH";"KERE FRANCIS";"ODONGO HARON"];

Reg_No=["BU/UP/2025/1531";"BU/UP/2025/1532";"BU/UP/2025/2109";"BU/UP/2025/2113";"BU/UP/2025/2142";"BU/UP/2025/3265";"BU/UP/2025/2110";"BU/UP/2025/5144";"BU/UP/2025/1560";"BU/UP/2025/1562";"BU/UP/2025/3215";"BU/UP/2025/4546";"BU/UP/2025/2114";"BU/UP/2025/2147";"BU/UG/2025/4443";"BU/UP/2025/2105";"BU/UG/2025/4444";"BU/UP/2025/1536";"BU/UP/2025/1539";"BU/UG/2025/4442";"BU/UG/2025/4440";"BU/UP/2025/3260";"BU/UG/2025/0013";"BU/UP/2025/1559";"BU/UP/2025/1561";"BU/UP/2025/1538";"BU/UP/2025/1541";"BU/UP/2025/4656";"BU/UP/2025/5146";"BU/UP/2025/2126";"BU/UP/2025/2137";"BU/UP/2025/3281";"BU/UP/2025/4629";"BU/UP/2025/1550";"BU/UP/2025/2100";"BU/UP/2025/3251";"BU/UP/2025/1546";"BU/UP/2025/2127";"BU/UP/2025/2138";"BU/UP/2025/5147";"BU/X

```
Hostel=["NJUKI";"NJUKI";"NJUKI";"NJUKI";"NJUKI";"NYERERE";"NYERERE";"NYERERE";"NYERERE";"NYERERE";"NYERERE";"LOGIN";"LOGIN";"LOGIN";"LOGIN";"LOGIN";"LOGIN";"PLUMBERS";"PLUMBERS";"
```


NO.1B

```
StudentsinMyProgram = readtable('C:\Users\essie\OneDrive\Desktop\GROUP 11 WAR\WAR 2  
MATLAB.xlsx')
```

NO.1C

writetable(StudentsinMyProgram,"GROUP 11 WAR.xlsx")

NO.1D

```
bar(Name, Age, 'g');
```

```
title('A graph of Name against Age');
```

```
xlabel("Names of students");
```

```
ylabel("Age")
```

```
save
```

NO.2

% Program to Calculate CPA and CGPA

% total number of semesters

num_semesters = input('Enter the total number of semesters: ');

total_points_all_sem = 0;

total_credits_all_sem = 0;

% Loop through each semester

for sem = 1:num_semesters

 fprintf('\n- Semester %d -\n', sem);

 num_courses = input('Enter the number of courses in this semester: ');

 sem_points = 0;

 sem_credits = 0;

% Loop through each course in the current semester

for course = 1:num_courses

 fprintf(' Course %d:\n', course);

 credits = input(' Enter credit units: ');

 grade_point = input(' Enter grade point (e.g., 4.0, 3.5): ');

% Accumulate points and credits for the current semester

 sem_points = sem_points + (credits * grade_point);


```

        sem_credits = sem_credits + credits;
    end

    % Calculate CPA (Semester GPA)
    if sem_credits > 0
        cpa = sem_points / sem_credits;
        fprintf('Semester %d CPA: %.2f\n', sem, cpa);
    else
        fprintf('Semester %d CPA: 0.00 (No credits entered)\n', sem);
    end

    % Accumulate overall totals for CGPA
    total_points_all_sem = total_points_all_sem + sem_points;
    total_credits_all_sem = total_credits_all_sem + sem_credits;
end

% Calculate CGPA (Cumulative GPA)
fprintf('\n=\n');
if total_credits_all_sem > 0
    cgpa = total_points_all_sem / total_credits_all_sem;
    fprintf('Final Cumulative CGPA: %.2f\n', cgpa);
else
    fprintf('Final Cumulative CGPA: 0.00\n');
end
fprintf('=\n');

```